

Roll No.

TME-306

B. TECH. (MECHANICAL ENGINEERING)
(THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
ENGINEERING MECHANICS

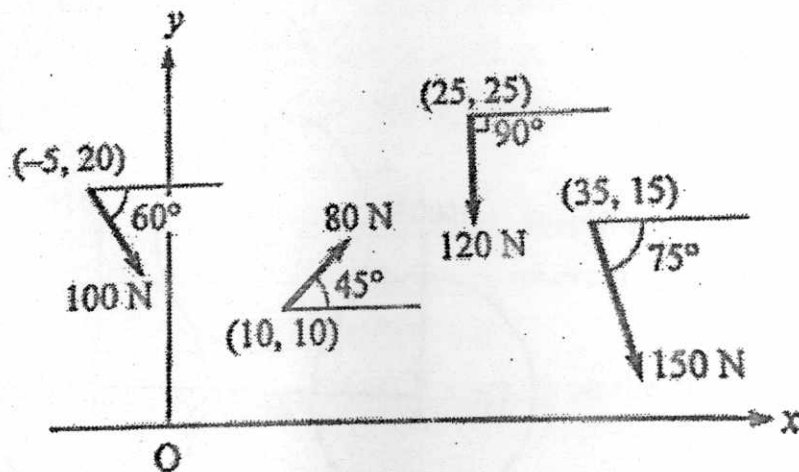
Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Determine the resultant of the force system shown in the figure. Assume that the coordinates of different points are in metres. (CO2)



P. T. O.

OR

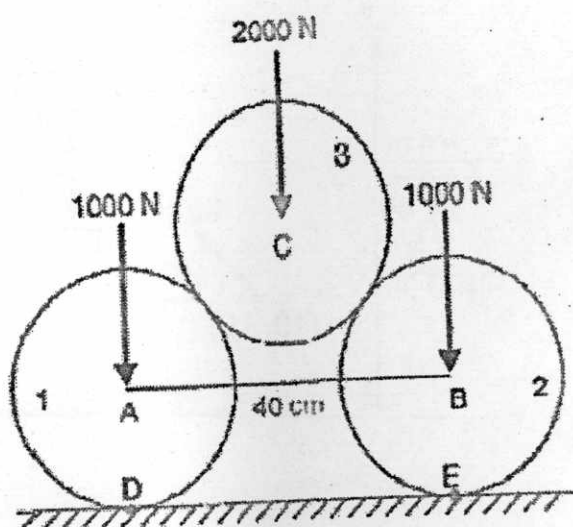
- (b) The resultant of two concurrent forces is 2500 N and the angle between the forces is 90° . The resultant makes an angle of 46° with one of the forces. Find the magnitude of each force. (CO2)
2. (a) What do you understand by the term 'Couple' ? Discuss the characteristics of a couple. (CO3)

OR

- (b) Define moment of force about a point and show that the algebraic sum of the moments of two coplanar forces about a point is equal to the moment of their resultant about that point. (CO3)
3. (a) Define a Free Body Diagram. Give *two* examples. State the conditions for equilibrium of a rigid body in three dimensions. (CO1)

OR

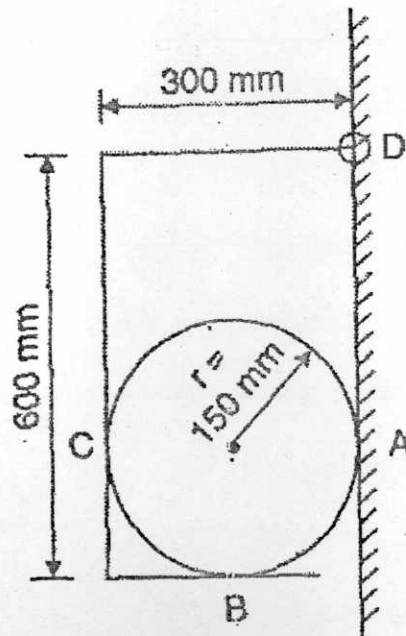
- (b) Classify the various kinds of system of forces with illustrative sketches. (CO1, CO3)
4. (a) Two smooth circular cylinders, each of weight $W = 1000$ N and radius 15 cm, connected at their centres by a string AB of length 40 cm and rest upon a horizontal plane supporting above them a third cylinder of weight 2000 N and radius 15 cm as shown in fig. below. Find the force in string AB and the pressure produced on the floor at the points of contact D and E. (CO3)



(3)

OR

- (b) A 600 N cylinder is supported by the frame ABCD as shown in given figure. The frame is hinged at D. Determine the reactions developed at contact points A, B, C and D. Neglect the weight of frame and assume all contact surfaces are smooth. (CO3)



5. (a) A drone is being directed across a frictionless ice-covered lake. The mass of the drone is 1.50 kg and its velocity is $3.00 \mathbf{i}$ m/s. After 10.0 s, the velocity is $9.00 \mathbf{i} + 4.00 \mathbf{j}$ m/s. If a constant force in the horizontal direction is causing this change in motion, find (i) the components of the force and (ii) the magnitude of the force. (CO1)

OR

- (b) A body of mass m moves in a horizontal direction such that at time t its position is given by $x(t) = at^4 + bt^3 + ct$, where a , b and c are constants.
- (i) What is the acceleration of the body?
- (ii) What is the time-dependent force acting on the body? (CO1)